Devam Zanzmera

AI/ML Engineer

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SUMMARY

- AI/ML Engineer with 3+ years of production experience building and deploying machine learning models with SHAP explainability for healthcare and banking, deployed via FastAPI/Docker on AWS and Azure.
- Beyond core production work, built independent projects spanning NLP and retrieval (LangChain, Hugging Face embeddings, ChromaDB), real-time data pipelines (Kafka, BERT-based multilingual sentiment analysis), and computer vision (YOLOv8 object detection and tracking).
- Comfortable working across the ML lifecycle from data pipelines and feature engineering to model evaluation, explainability, and production deployment.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Bash
Data Analytics: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI
ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers
Computer Vision: YOLOv8, Object Detection, Object Tracking, Roboflow
Generative AI & NLP: GPT-4, LLaMA, BERT, RAG, Prompt Engineering, Agentic AI
LLM Ops & Tooling: LangChain, LangGraph, Pinecone, FAISS, Chroma DB, Vector Databases
ML Ops & Deployment: MLflow, Docker, Kubernetes, GitHub Actions, CI/CD, FastAPI, Django
Cloud: AWS (SageMaker, Lambda, EC2), Azure (ML Studio, Databricks)
Big Data & Pipelines: Databricks, Apache Spark, PySpark, Kafka, Airflow, Feature Engineering
Databases: MySQL, PostgreSQL, SQL Server, MongoDB, Elasticsearch
Model Evaluation: SHAP, ROC-AUC, F1 Score, Precision, Recall, Cross-Validation

EXPERIENCE

AI/ML Engineer | Oracle Health | Toronto, Canada | Contract 06/2024 – Present

- Developed patient risk stratification system for Oracle Health Canada to accurately predict hospital readmissions across three Canadian provinces, enabling early clinical intervention for high-risk patients.
- Engineered PIPEDA-compliant data pipelines using Databricks to extract, validate, and process patient EHR records from legacy Oracle systems.
- Built feature engineering process using PySpark to transform raw clinical EHR data into 50+ validated features including medication interactions and comorbidity scoring.
- Conducted comparative analysis using Scikit-learn to evaluate Logistic Regression and Random Forest baseline algorithms, identifying XGBoost as optimal solution.
- Developed XGBoost production model achieving 85% AUC performance, utilizing MLflow for experiment tracking across 50+ model iterations.
- Implemented SHAP explainability framework to generate patient-specific risk factor visualizations for clinical teams.
- Deployed production model to AWS SageMaker with automated GitHub Actions CI/CD pipeline, establishing monitoring for data drift detection.
- Implemented a hybrid inference architecture using AWS SageMaker, supporting scheduled batch inference for population-level patient risk scoring and real-time inference for time-sensitive clinical cases.
- Built FastAPI microservice to serve real-time risk predictions with token-based authentication and encrypted payload handling for PIPEDA compliance.

AI/ML Engineer | Accenture | Mumbai, India | Permanent 09/2021 – 04/2023

- Supported credit risk stratification initiative for Accenture BFSI consulting practice to predict loan defaults, targeting a 20% reduction in default losses.
- Executed SQL queries to extract, clean, and validate loan records from legacy banking core systems.
- Implemented robust data cleaning pipeline using Pandas to handle missing values, outliers, and formatting inconsistencies.
- Created 50 domain-specific financial features using NumPy including debt-to-income ratio and credit score velocity.

Utilized Scikit-learn and XGBoost for default prediction model development with MLflow experiment tracking, improving performance by 15% over logistic regression benchmarks.

- Executed cross-validation framework across five-fold stratified splits, achieving 74% ROC-AUC on held-out test data.
- Implemented SHAP explainability framework generating customer-specific risk factor visualizations for credit officers.
- Created FastAPI microservice containerized via Docker and deployed on Azure App Service, exposing real-time loan risk scoring endpoint with sub-100ms latency.
- Delivered model operationalization support through PostgreSQL database optimization, reducing credit review effort by 30%.

PROJECTS

E-Commerce Support RAG Agent

- Engineered RAG-based e-commerce customer service AI agent POC using LangChain and ChromaDB, grounding responses in verified policy documents to reduce hallucinations.
- Implemented semantic search leveraging Hugging Face open-source embeddings to automatically surface support guidelines.
- Prototyped an interactive end-to-end interface using Streamlit, designing the complete POC as demo-ready for non-technical stakeholders.

EDUCATION

Master of Applied Computing — Specialization in Artificial Intelligence
University of Windsor, Windsor, ON, Canada | 05/2023 – 08/2024

Bachelor of Technology in Computer Engineering
Pandit Deendayal Energy University, Gandhinagar, Gujarat, India | 08/2018 – 05/2022